



DATA PREPROCESSING REPORT

Machine Learning & Deep Learning Prediction Project

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Project Title:

AI-Powered Early Pediatric Pneumonia Detection for kids& EMR Integration in Algeria

February 28, 2026

GitHub

<https://github.com/AminaMar/pediatric-pneumonia-detection> CNN

Prediction Target: Pneumonia (Yes/No) — Binary classification from clinical vitals (CSV) and chest X-ray images

Task Type: ☒ **Binary Classification** ☐ Multi-class ☐ Regression

SECTION 1 — Dataset Overview & Exploration

1.1 Dataset Description

Dataset name / source:

1 — Branch A · CSV Clinical: Pneumonia Patient Dataset by essienmary · Kaggle

URL: <https://www.kaggle.com/datasets/essienmary/pneumonia-dataset>

Type: Structured CSV — patient vitals, symptoms, diagnosis label (Yes/No)

2 — Branch B · X-Ray Images: Chest X-Ray Images (Pneumonia) — Kermany et al., 2018 · Kaggle

URL: <https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>

Type: JPEG chest X-ray images — NORMAL / PNEUMONIA — pediatric patients aged 1–5

Why is this dataset appropriate for your prediction target?

1 — CSV: Contains the exact clinical indicators used to diagnose pneumonia in a hospital setting — fever, SpO₂, fatigue, cough, crackles, temperature. After filtering to pediatric ages 1–16, it provides a structured clinical signal. Structured clinical data + X-ray together mirror how a real physician diagnoses pediatric pneumonia.

2 — X-Ray: 5,856 pediatric chest X-rays (ages 1–5) graded by two expert physicians with third-party verification. Standard benchmark for pediatric pneumonia AI (CheXNet trained on this). Directly relevant to our Algerian hospital deployment where specialist radiologists are scarce.

1.2 Dataset Statistics

Metric	Branch A — CSV	Branch B — X-Ray	Notes
Total samples (raw)	710 patients	5,856 images	CSV contains all age groups → must filter
Samples after pediatric filter	127 patients (ages 1–16)	5,856 images (ages 1–5)	Age filter applied only to CSV
Number of features	12 clinical features	224×224×1 pixel array	CSV: dropped PatientID, Xray col, WBC_count
Target classes	0=Healthy (73, 57.5%) 1=Pneumonia (54, 42.5%)	0=NORMAL (1,583, 27%) 1=PNEUMONIA (4,273, 73%)	Binary classification — both branches
File format	CSV (.csv)	JPEG (.jpeg)	—
Dataset size	~50 KB	> 1 GB	X-ray requires Google Drive / Colab
Missing values	2,988 NaN raw; Xray 96.9%, WBC 96.9%	None — image files complete	Two CSV columns dropped; rest imputed

1.3 Feature Inventory

Branch A — CSV Clinical (12 features after dropping 3):

- Fatigue** (None/Mild/Moderate/Severe) **importance=0.6481** — MOST IMPORTANT — ~8% missing → 'Unknown' + LabelEncoded
- Fever** (None/Low/High) **importance=0.1290** — 48% missing → 'Unknown' + LabelEncoded
- Temperature** (Numeric °C) **importance=0.0919** — ~8% missing → SimpleImputer(median)
- Chest_pain** (None/Mild/Severe) **importance=0.0492** — 58.3% missing → 'Unknown'
- Cough** (None/Dry/Bloody) **importance=0.0340** — 49.6% missing → 'Unknown'
- Age** (Numeric 1–16 years) **importance=0.0318** — Used for pediatric filter. 0% missing
- Shortness_of_breath** (None/Mild/Severe) **importance=0.0063** — 52% missing → 'Unknown'
- Sputum_color** (Clear/Yellow/Green/Rust/None) **importance=0.0038** — 12.6% missing → 'Unknown'
- Oxygen_saturation** (Numeric % SpO₂) **importance=0.0030** — 10.2% missing → median
- Confusion** (Yes/No) **importance=0.0020** — 22% missing → 'Unknown'
- Gender** (M/F → 1/0) **importance=0.0005** — 0% missing. LabelEncoded
- Crackles** (Yes/No (lung auscultation)) **importance=0.0003** — ~5% missing → 'Unknown'

DROPPED: PatientID (ID only) | Xray col (96.9% missing) | WBC_count (96.9% missing)

Branch B — X-Ray:

Grayscale chest radiograph → resized to 224×224 px → normalized to [0.0, 1.0] float32. Input shape: (224, 224, 1). Validated: Min=0.0000, Max=0.9961, Mean=0.5069, Std=0.2676.

1.4 EDA Summary

Finding	CSV Branch	X-Ray Branch
Temperature	Pneumonia mean 38.80°C vs Healthy 37.68°C — clear separation	Not applicable (image data)

Finding	CSV Branch	X-Ray Branch
SpO ₂	Pneumonia median 93.0% vs Healthy 94.0% — small diff; individual vitals insufficient alone	NORMAL ≈ darker pixels (air); PNEUMONIA ≈ brighter (fluid/consolidation)
Fatigue importance	0.6481 — dominates all other features combined	N/A — CNN learns features implicitly via convolution
Class balance	42.5% Pneumonia / 57.5% Healthy — moderate imbalance	73% PNEUMONIA / 27% NORMAL — severe 2.70:1 imbalance
Visualizations	13 charts: feature importance bar, correlation heatmap, boxplots by diagnosis, age dist, symptom rates, ROC curves, confusion matrices, CV violin	Dataset split comparison, class balance bar charts, sample image grids

Most interesting EDA finding:

CSV: Fatigue importance = 0.6481, far exceeding Fever (0.129) and Temperature (0.092) combined. Clinically significant — how tired a child appears is more informative than measured vitals. Supports the multimodal approach.

X-Ray: Original Kaggle validation set = only 16 images (8+8). One misclassification = 6.25% accuracy swing — completely unreliable for selecting among 8 CNN architectures. Fixed by rebuilding to 70/15/15 stratified split → val: 878 images.

SECTION 2 — Data Quality Issues Identified

2.1 Issues Checklist

Issue	Present?	Severity	Details / Count	Fix Applied
Missing values	✓ Yes	✓ High	CSV: 2,988 NaN. Chest_pain 58.3%, SOB 52%, Cough 49.6%, Fever 48%, Confusion 22%, Xray col 96.9%, WBC 96.9%.	Categorical NaN → fillna('Unknown') + LabelEncoder. Numeric → SimpleImputer(median). Sparse cols → dropped.
Class imbalance	✓ Yes	✓ High	X-Ray: PNEUMONIA 73% vs NORMAL 27%, ratio 2.70:1. CSV: 42.5% Pneumonia vs 57.5% Healthy.	X-Ray: class_weight {0:1.850, 1:0.685} + augmentation. CSV: class_weight='balanced' inside Pipeline.
Duplicate records	✓ No	N/A	0 duplicates found.	df.drop_duplicates() confirmed 0 removed.
Outliers	✓ Yes	✓ Med	Temperature: few >40°C. SpO ₂ : few <85%. Clinically real extreme values.	No removal — extremes are valid disease presentations. Median imputation is robust.
Inconsistent format	✓ Yes	✓ High	CSV uses '-' for missing data instead of NaN standard.	df.replace('-', np.nan) applied at load time before any processing.

Issue	Present?	Severity	Details / Count	Fix Applied
Variable image dims	☑ Yes	☑ High	X-Ray original images have variable resolutions from different hospital devices.	Resize all to 224×224 px using cv2.resize with INTER_AREA interpolation.
Non-normalized values	☑ Yes	☑ High	JPEG pixels [0,255] uint8. Clinical numerics at different scales (°C, %, bpm).	X-Ray: ÷255.0 → [0,1] float32. CSV: StandardScaler inside Pipeline (fit on train fold only).
Label noise / errors	☑ Yes	☐ Low	~5–10% estimated label noise inherent in any medical image dataset.	Kermany protocol: graded by 2 physicians + third for verification. Accepted as-is.
Age mismatch (CRITICAL)	☑ Yes	☑ High	CSV ages 3–90. X-Ray ages 1–5 only. Mixing creates clinically invalid cross-modal patterns.	Filtered CSV to ages 1–16 → 710 → 127 pediatric patients. Mandatory before any training.
Inadequate val split	☑ Yes	☑ High	Kaggle X-Ray val = 16 images (8+8). One error = 6.25% swing — unreliable for model selection.	Rebuilt: stratified 70/15/15 → Train:4,099 / Val:878 / Test:879 images.

2.2 Most Critical Issues

Issue 1 — Age Mismatch (CRITICAL):

The X-ray dataset contains ONLY pediatric patients aged 1–5. The CSV dataset originally contained ages 3–90. A multimodal AI trained on mixed-age clinical data and pediatric-only X-rays would learn clinically incorrect cross-modal relationships. Vital sign ranges, typical pathogens, and X-ray patterns differ fundamentally between a 2-year-old and a 70-year-old.

Fix: Filtered CSV to ages 1–16 → 127 pediatric patients. Mandatory step before any training.

Issue 2 — Kaggle Validation Set = 16 Images (CRITICAL):

Training 8 CNN architectures with early stopping using 16 validation images produces completely unreliable metrics. One misclassified image = 6.25% accuracy swing. Early stopping and model selection are meaningless at this size.

Fix: Rebuilt entire split: stratified 70/15/15 — validation set grew from 16 → 878 images.

SECTION 3 — Dataset Splitting Strategy

3.1 Branch A — CSV (127 patients) ☑ Training COMPLETE

Split	% Total	Samples	Healthy (0)	Pneumonia (1)	Notes
Training	75%	95 patients	55 (57.9%)	40 (42.1%)	5-fold CV: ~76 train / ~19 val per fold
Validation	Dynamic — inside 5-fold CV	~19 per fold	~11 per fold	~8 per fold	No fixed val set. CV creates rotating folds from X_train (95 patients) small data size Advanced Validation — 3 Approaches (Bootstrap Validation

					(1 000 iterations)- Fold Stratified Cross-Validation- Repeated Stratified K-Fold (10 x 5 = 50 evaluations)
Split	% Total	Samples	Healthy (0)	Pneumonia (1)	Notes
Test	25%	32 patients	18 (56.3%)	14 (43.7%)	Final evaluation — never touched during training
TOTAL	100%	127 patients	73 (57.5%)	54 (42.5%)	Ratio 1.35:1 — moderate imbalance

3.1 Branch B — X-Ray (5,856 images) ⌚ Training Pending

Split	% Total	Samples	NORMAL (0)	PNEUMONIA (1)	Notes
Training	70%	4,099 images	1,108 (27.0%)	2,991 (73.0%)	Augmentation applied here only
Validation	15%	878 images	237 (27.0%)	641 (73.0%)	FIXED — early stopping & model selection. Was 16 images in Kaggle → REBUILT.
Test	15%	879 images	238 (27.1%)	641 (72.9%)	Final held-out — never touched
TOTAL	100%	5,856 images	1,583 (27%)	4,273 (73%)	Ratio 2.70:1 — severe imbalance → class weights required

☒ Did you use stratified splitting? ☒ Yes ☐ No ☐ Not applicable Explain why: Branch A — CSV: train_test_split(stratify=y, random_state=42) preserves the 42.5%/57.5% pneumonia/healthy ratio in both train (95 patients) and test (32 patients). StratifiedKFold(n_splits=5) applied inside cross-validation on X_train. Branch B — X-Ray: sklearn train_test_split(stratify=labels) applied twice for 70/15/15, preserving the 27%/73% NORMAL/PNEUMONIA ratio across all three splits. Essential given the 2.70:1 class imbalance.	☐ Original validation set issue? ☒ Too small ☐ Adequate ☐ Non-existent Action taken: Branch A — CSV: No fixed validation set. With n=127 patients, a dedicated val set would be statistically unreliable. Solution: 5-Fold Stratified Cross-Validation on X_train (95 patients) creates ~19-patient rotating folds, giving robust AUC estimates across all 5 iterations. Branch B — X-Ray: CRITICAL — Kaggle original val = 16 images (8+8). One misclassification = 6.25% swing, making model selection across 8 CNNs unreliable. Action: Full 70/15/15 stratified re-split → validation grew from 16 → 878 images (+54×).
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SECTION 4 — Preprocessing Pipeline + Full Source Code

4.1 Image Preprocessing — Branch B (X-Ray)

Step	Applied?	Value / Method	Justification
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Resize images	✓ Yes	224×224 px — cv2.INTER_AREA	Standard ImageNet input for all 8 architectures. INTER_AREA best for downscaling.
Normalize pixel values	✓ Yes	[0,1] → ÷255.0 (all models) [-1,1] → ÷127.5-1 (MobileNetV2)	Stabilizes gradients. MobileNetV2 requires [-1,1] — mixing silently reduces accuracy 5–15%.
Step	Applied?	Value / Method	Justification
Convert color space	✓ Yes	RGB → Grayscale via cv2.IMREAD_GRAYSCALE	X-rays inherently grayscale. Shape: (224,224,1).
Convert to float32	✓ Yes	img.astype(np.float32)	Required by TensorFlow. float16 causes sigmoid overflow → NaN loss or 50% stuck accuracy.
Quality filtering	✓ Yes	Laplacian variance — NORMAL threshold <87.7, PNEUMONIA <62.3	Bottom 10% blurriest images identified. Optional removal estimated +0.5%.
Data augmentation (train only)	✓ Yes	Rotation ±15°, Shift 10%, Shear 10%, Zoom 10%, H-Flip=True, V-Flip=False, fill='nearest'	Prevents overfitting. V-Flip disabled — anatomically incorrect for chest X-rays.
Add channel dimension	✓ Yes	np.expand_dims(img, axis=-1) → (224,224,1)	TensorFlow requires explicit channel dim for grayscale input.

4.2 CSV Preprocessing — Branch A

Step	Applied?	Method / Value	Columns Affected
Replace '-' with NaN	✓ Yes	df.replace('-', np.nan)	All columns — raw CSV uses '-' for missing instead of NaN
Remove duplicates	✓ Yes	df.drop_duplicates()	Full dataset — 0 duplicates found
Pediatric filter	✓ Yes	df[df.Age.between(1,16)] → 710 → 127	Age column — critical for multimodal alignment
Drop sparse columns	✓ Yes	df.drop(['PatientID', 'Xray', 'WBC_count'])	3 columns: ID + 96.9% missing each
Fill categorical NaN	✓ Yes	col.fillna('Unknown')	Cough (49.6%), Fever (48%), SOB (52%), Chest_pain (58.3%), Confusion (22%), Sputum_color (12.6%), Fatigue (~8%), Crackles (~5%)
Encode categoricals	✓ Yes	LabelEncoder — text → integer	Gender, Cough, Fever, SOB, Chest_pain, Fatigue, Confusion, Crackles, Sputum_color
Impute numeric NaN	✓ Yes	✓ Median — SimpleImputer(strategy='median')	Oxygen_saturation (10.2%), Temperature (~8%)
Encode target	✓ Yes	(df['Diagnosis']=='Yes').astype(int)	Diagnosis: Yes→1, No→0

Feature scaling	☒ Yes	StandardScaler inside sklearn Pipeline (fit on X_train fold only — no leakage)	SVM, LR, KNN, MLP. RF/GB/NB: no scaling needed.
Train/Test split	☒ Yes	train_test_split(test_size=0.25, stratify=y, random_state=42)	75% → 95 train 25% → 32 test
5-Fold Stratified CV	☒ Yes	StratifiedKFold(n_splits=5, shuffle=True, random_state=42)	Applied to X_train (95 patients) — rotating val fold ~19 patients per fold

4.3 Complete Source Code — Batch A (CSV Branch)

All code is extracted exactly from pediatric_pneumonia_analysis_COLAB.ipynb — 49 cells, runs smoothly end-to-end on Google Colab with zero errors.

STEP 4: Preprocessing Pipeline (Replace NaN → Deduplicate → Filter → Encode → Impute)

```
# — STEP 4: Preprocessing Pipeline —————
# Load CSV → Replace '-'→NaN → Deduplicate → Filter Age 1-16
# → Convert strings → Analyze missing → Encode → Impute → Encode target

# 4.1 Replace dash '-' with NaN ('-' = measurement not taken)
df.replace('-', np.nan, inplace=True)
print(f"☑ All '-' replaced with NaN")
print(f'    Total NaN values: {df.isnull().sum().sum()}')

# 4.2 Remove duplicate rows
before = len(df)
df.drop_duplicates(inplace=True)
print(f'Duplicate rows removed: {before - len(df)}')

# 4.3 Filter: keep ONLY pediatric patients (age 1-16)
# .between(1,16) is inclusive. .copy() avoids SettingWithCopyWarning
kids = df[df['Age'].between(1, 16)].copy()
print(f'Full dataset : {len(df):4d} patients')
print(f'Pediatric only: {len(kids):4d} patients (ages 1-16)')
counts = kids['Diagnosis'].value_counts()
for label, n in counts.items():
    pct = n / len(kids) * 100
    print(f' {label}: {n:3d} ({pct:.1f}%)')

# 4.4 Convert string columns to proper numeric type
# errors='coerce' → non-numeric values become NaN instead of crashing
kids['Oxygen_saturation'] = pd.to_numeric(kids['Oxygen_saturation'], errors='coerce')
kids['WBC_count']         = pd.to_numeric(kids['WBC_count'],         errors='coerce')
print('☑ Numeric conversion complete')

# 4.5 Missing value analysis
missing_count = kids.isnull().sum()
missing_pct   = (missing_count / len(kids) * 100).round(1)
miss_df = pd.DataFrame({'Missing Count': missing_count,
                        'Missing %':      missing_pct}).sort_values('Missing %',
ascending=False)
strat_map = {
    'Xray':          '✗ DROP  - 96.9% missing',
    'WBC_count':     '✗ DROP  - 96.9% missing',
    'Chest_pain':    'G  FILL  → Unknown',
    'Shortness_of_breath': 'G  FILL  → Unknown',
    'Cough':         'G  FILL  → Unknown',
    'Fever':         'G  FILL  → Unknown',
    'Confusion':     'G  FILL  → Unknown',
    'Sputum_color':  'G  FILL  → Unknown',
    'Fatigue':       'G  FILL  → Unknown',
    'Oxygen_saturation': '👉 FILL  → median',
    'Temperature':   '👉 FILL  → median',
}
print(miss_df[miss_df['Missing Count'] > 0].to_string())
```

STEP 6.1: Select 12 Clinical Features

```
# — STEP 6.1: Select 12 clinical features —————
# DROPPED: PatientID (ID only), Xray (96.9% missing), WBC_count (96.9%)

FEATURES = [
    'Gender',          # Categorical    — M / F
    'Age',             # Numeric      — 1-16 years
    'Cough',           # Categorical  — None / Dry / Bloody
    'Fever',           # Categorical  — None / Low / High
    'Shortness_of_breath', # Categorical — None / Mild / Severe
    'Chest_pain',      # Categorical  — None / Mild / Severe
    'Fatigue',         # Categorical  — None / Mild / Moderate / Severe
    'Confusion',       # Categorical  — Yes / No
    'Oxygen_saturation', # Numeric      — SpO2 85-100%
    'Crackles',        # Categorical  — Yes / No (lung auscultation)
    'Sputum_color',    # Categorical  — Clear/Yellow/Green/Rust/None
    'Temperature',     # Numeric      — body temp in °C
]
TARGET = 'Diagnosis' # Yes=Pneumonia (1), No=Healthy (0)
print(f'✅ {len(FEATURES)} features selected, target: {TARGET}')
```

STEP 6.2: Encode Categoricals + Impute Missing Values

```
# — STEP 6.2: Encode categoricals + impute missing —————
model_df = kids[FEATURES + [TARGET]].copy()

# CATEGORICAL: fill NaN → 'Unknown', then LabelEncode (text → integer)
# LabelEncoder maps: Female→0, Male→1 / No→0, Yes→1 etc.
CATEGORICAL_COLS = [
    'Gender', 'Cough', 'Fever', 'Shortness_of_breath',
    'Chest_pain', 'Fatigue', 'Confusion', 'Crackles', 'Sputum_color'
]
le = LabelEncoder()
for col in CATEGORICAL_COLS:
    model_df[col] = model_df[col].fillna('Unknown')
    model_df[col] = le.fit_transform(model_df[col].astype(str))

# NUMERIC: impute remaining NaN with median (robust to outliers)
imputer = SimpleImputer(strategy='median')
model_df[FEATURES] = imputer.fit_transform(model_df[FEATURES])

# TARGET: encode Yes→1 (Pneumonia), No→0 (Healthy)
model_df[TARGET] = (model_df[TARGET] == 'Yes').astype(int)

print('✅ Encoding and imputation complete')
print(f'   Shape: {model_df.shape}')
print(f'   Missing values remaining: {model_df.isnull().sum().sum()}')
print(f'   Class balance: {dict(model_df[TARGET].value_counts())}')
model_df.head()
```

STEP 6.3: Train / Test Split (75% / 25%, stratified)

```
# — STEP 6.3: Train / Test split (75% / 25%) —————
X = model_df[FEATURES].values # Feature matrix (numpy array)
y = model_df[TARGET].values   # Target vector

# stratify=y → ensures proportional class distribution in both sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.25,      # 25% → 32 patients
    random_state=42,     # reproducibility
    stratify=y           # preserve 42.5%/57.5% ratio in each set
)

print(f'Training samples : {len(X_train)} (75%)')
print(f'Test samples      : {len(X_test)} (25%)')
unique, counts = np.unique(y_train, return_counts=True)
```

```
for u, c in zip(unique, counts):
    label = 'Pneumonia' if u==1 else 'Healthy '
    print(f' {label}: {c} ({c/len(y_train)*100:.1f}%)')
```

4.3 Complete Source Code — Batch B (X-Ray Branch)

All code extracted from `02_Data_Preprocessing.ipynb` + `03_Data_Loaders.ipynb` — runs on Google Colab GPU with zero errors.

STEP 1: Reorganize Dataset — Stratified 70/15/15 Split

```
# Original Kaggle val = 16 images (8+8) → unreliable for 8 CNN architectures
import os, shutil
from sklearn.model_selection import train_test_split
# Collect all paths + labels from train/val/test folders
all_images, all_labels = [], []
for split in ['train', 'val', 'test']:
    for cls, lbl in [('NORMAL', 0), ('PNEUMONIA', 1)]:
        for img in os.listdir(f'{RAW_PATH}/{split}/{cls}'):
            all_images.append(f'{RAW_PATH}/{split}/{cls}/{img}')
            all_labels.append(lbl)
# Stratified 70/15/15 split — preserves 27%/73% class ratio
X_tr, X_tmp, y_tr, y_tmp = train_test_split(all_images, all_labels, test_size=0.30, stratify=all_labels, random_state=42)
X_val, X_test, y_val, y_test = train_test_split(X_tmp, y_tmp, test_size=0.50, stratify=y_tmp, random_state=42)
print(f'✅ Train:{len(X_tr)} / Val:{len(X_val)} / Test:{len(X_test)}' # 4099 / 878 / 879)
```

STEP 2: Image Preprocessing (Grayscale → Resize 224×224 → Normalize → float32)

```
import cv2, numpy as np
def preprocess_image(img_path, img_size=224):
    # 1. Load as grayscale (X-rays are inherently grayscale)
    img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
    # 2. Resize 224x224 using INTER_AREA (best for downscaling)
    img = cv2.resize(img, (img_size, img_size), interpolation=cv2.INTER_AREA)
    # 3. Normalize [0,255] → [0,1] float32 (prevents sigmoid overflow)
    img = img.astype(np.float32) / 255.0
    # 4. Add channel dim → (224, 224, 1) required by TensorFlow
    return np.expand_dims(img, axis=-1) # shape: (224, 224, 1)
```

STEP 3: TensorFlow Data Generators + Augmentation + Class Weights

```
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from sklearn.utils.class_weight import compute_class_weight
# Training generator WITH augmentation
train_gen = ImageDataGenerator(rescale=1./255, rotation_range=15,
    width_shift_range=0.1, height_shift_range=0.1, shear_range=0.1,
    zoom_range=0.1, horizontal_flip=True,
    vertical_flip=False, # DISABLED: heart position matters
    fill_mode='nearest')
# Val/Test: normalization ONLY (no augmentation)
val_test_gen = ImageDataGenerator(rescale=1./255)
# Class weights to compensate 2.70:1 imbalance
cw = compute_class_weight('balanced', classes=np.array([0,1]), y=y_tr)
class_weights = {0: cw[0], 1: cw[1]} # {0: 1.850, 1: 0.685}
# Validation: batch shape (32, 224, 224, 1) ✅ | pixel range [0,1] ✅ | float32 ✅
print(f'✅ class_weights: {class_weights}' # {0: 1.850, 1: 0.685})
```

SECTION 5 — Class Imbalance Handling

5.1 Imbalance Analysis

Branch	Class / Label	Count	Percentage	Notes
CSV — Branch A	Class 0 / Healthy	73	57.5%	Slightly majority
CSV — Branch A	Class 1 / Pneumonia	54	42.5%	Minority — target
CSV — TOTAL		127	100%	Ratio 1.35:1 — moderate
X-Ray — Branch B	Class 0 / NORMAL	1,583	27.0%	Strongly under-represented
X-Ray — Branch B	Class 1 / PNEUMONIA	4,273	73.0%	Dominant class
X-Ray — TOTAL		5,856	100%	Ratio 2.70:1 — severe

5.2 Strategy Chosen

Strategy	Applied?	Details / Parameters	Expected Impact
Class weights (recommended)	☒ Yes — BOTH	CSV: class_weight='balanced' in RF, SVM, LR (auto-computed). X-Ray: class_weight={0:1.850, 1:0.685} passed to model.fit()	Compensates imbalance without modifying data. Zero risk of generating invalid samples.
Data augmentation (train only)	☒ Yes — X-Ray	Rotation ±15°, Shifts 10%, Shear 10%, Zoom 10%, H-Flip=True, V-Flip=False. Keras ImageDataGenerator on training set ONLY.	Increases diversity 50–100%. Val/Test: normalization only.
CSV Normalization (StandardScaler)	☒ Yes — CSV	StandardScaler inside sklearn Pipeline — fit ONLY on X_train fold. Applied to: Temperature, Oxygen_saturation, Age (numeric features). RF and GB excluded (tree-based, scale-invariant).	Guarantees zero data leakage in 5-fold CV. SVM, LR, KNN, MLP require normalization for equal feature weighting.
Oversampling — SMOTE	☐ No	SMOTE cannot synthesize chest X-ray pixels. For CSV: biologically implausible with n=95.	Rejected — class weights achieve same correction with zero risk.
Undersampling	☐ No	Would remove NORMAL X-rays — already strongly under-represented.	Rejected — would worsen imbalance.

5.3 Class Weight Calculation

Formula: $\text{weight_c} = \text{n_samples} / (\text{n_classes} \times \text{n_samples_c})$

X-Ray branch (train set: 4,099 images):

weight_NORMAL (0) = $4,099 / (2 \times 1,108) = \mathbf{1.850}$

weight_PNEUMONIA (1) = $4,099 / (2 \times 2,991) = \mathbf{0.685}$

CSV branch — sklearn auto-computes with class_weight='balanced':

weight_Healthy (0) = $127 / (2 \times 73) = 0.870$

weight_Pneumonia (1) = $127 / (2 \times 54) = 1.176$

5.4 Code Snippet — Class Weights + Augmentation + Pipeline

```
# — X-RAY: Compute class weights —————
from sklearn.utils.class_weight import compute_class_weight
cw = compute_class_weight('balanced', classes=np.array([0,1]), y=y_train_labels)
class_weights = {0: cw[0], 1: cw[1]} # {0: 1.850, 1: 0.685}

# — X-RAY: Augmentation - training set ONLY —————
train_gen = ImageDataGenerator(rescale=1./255,
                               rotation_range=15, width_shift_range=0.1, height_shift_range=0.1,
                               shear_range=0.1, zoom_range=0.1,
                               horizontal_flip=True, vertical_flip=False, # V-Flip disabled (anatomy)
                               fill_mode='nearest')
val_gen = ImageDataGenerator(rescale=1./255) # NO augmentation on val/test

history = model.fit(train_data, epochs=25, validation_data=val_data,
                    class_weight=class_weights,
                    callbacks=[early_stop, checkpoint, reduce_lr])

# — CSV: sklearn Pipeline (prevents data leakage) —————
# StandardScaler is fit ONLY on train fold - never on validation data
from sklearn.pipeline import Pipeline
pipe = Pipeline([('scaler', StandardScaler()),
                 ('clf', RandomForestClassifier(class_weight='balanced'))])
cv_scores = cross_val_score(pipe, X_train, y_train, cv=skf, scoring='roc_auc')
print(f'CV AUC: {cv_scores.mean():.4f} ± {cv_scores.std():.4f}')
```

SECTION 6 — Critical Reflection & Next Steps

6.1 Methodology Justification

Strongest aspect of the preprocessing pipeline:

CSV: The sklearn Pipeline architecture guarantees zero data leakage — the StandardScaler is fit only on the training fold in each CV iteration. This is the single most critical correctness requirement in the entire pipeline. Without it, all AUC estimates would be inflated and non-reproducible.

X-Ray: Fixing the validation set from 16 → 878 images is the most impactful single step. Without this, no reliable early stopping, model selection, or hyperparameter comparison is possible across 8 CNN architectures.

What would you do differently with more time or data:

Branch A — CSV:

1. Tighten CSV age filter to ages 1–5 (exact match with X-ray dataset) if sample size permits after filtering.
2. Test SMOTE inside CV folds only (imblearn Pipeline) to assess recall improvement on the minority class (n=54 pneumonia).
3. Apply MICE (Multiple Imputation by Chained Equations) instead of 'Unknown' category for high-missingness features (Cough 49.6%, SOB 52%).

4. Collect more clinical data from Algerian hospitals to bring CSV n to ≥ 500 pediatric patients for reliable deep learning fusion.
- Branch B — X-Ray:**
1. Apply CLAHE (Contrast Limited Adaptive Histogram Equalization) to X-ray images for improved opacity visibility in consolidation regions.
 2. Add brightness variation augmentation ($\pm 15\%$) to simulate different X-ray exposure settings across Algerian hospital devices.
 3. Combine Kermany + NIH ChestX-ray14 + CheXpert datasets for a more generalised multi-source model.
 4. Remove the bottom 10% blurriest images (Laplacian variance threshold identified in EDA) to measure accuracy gain.

Which preprocessing decision are you least confident about? Why?

Branch A — CSV:

The ‘Unknown’ category for high-missingness categoricals (Cough 49.6%, SOB 52%, Fever 48%). ‘Unknown’ conflates two clinically different situations: symptom absent vs. symptom not recorded. A child with no cough is clinically different from one whose cough status was undocumented. MICE (Multiple Imputation by Chained Equations) would be preferred with more time.

Branch B — X-Ray:

The INTER_AREA interpolation choice for resizing from variable resolutions (384×127 to 2,916×2,713) down to 224×224. While INTER_AREA is theoretically optimal for downscaling, we have not verified whether fine diagnostic features (subtle interstitial patterns, early consolidation) survive the aggressive downscaling from images averaging 1,500×1,200 pixels. A formal ablation comparing INTER_AREA vs. INTER_LANCZOS4 at multiple output sizes (224, 299, 384) would confirm this choice.

6.2 Evaluation Metrics Planned

Metric	Use?	Why it matters for pneumonia detection	CSV Status	X-Ray Status
Accuracy	✔ Yes	Overall rate — reported alongside others. Misleading alone with imbalance.	✔ 93.75% (GB)	⌚ Pending
AUC-ROC (Primary)	✔ Yes	PRIMARY METRIC. Invariant to class imbalance. Discriminative power across all thresholds.	✔ 0.9960 (GB best)	⌚ Pending
F1-Score	✔ Yes	Harmonic balance of Precision/Recall. Penalizes both FP and FN.	✔ 0.9231 (GB)	⌚ Pending
Sensitivity / Recall	✔ Yes — CRITICAL	MOST CRITICAL: missed pneumonia in a child can be fatal. Target: >95%.	⚠ 0.8571 — below 95% target	⌚ Pending
Precision	✔ Yes	% of predicted pneumonia truly positive. Avoids alarm fatigue.	✔ 1.0000 (perfect)	⌚ Pending
5-Fold CV AUC	✔ Yes	Generalization reliability — essential for n=127.	✔ 0.9939±0.009	N/A — fixed val

Metric	Use?	Why it matters for pneumonia detection	CSV Status	X-Ray Status
Confusion Matrix	✓ Yes	Full TP/TN/FP/FN breakdown for clinical error analysis.	✓ All 7 models	⌚ Pending
PR-AUC	✓ Yes	More informative than ROC-AUC under strong imbalance (X-Ray 2.70:1).	✓ Computed	⌚ Pending

6.3 Next Steps

Task	Status	Target Date	Notes
Preprocessing — CSV	✓ DONE	Feb 2026	0 NaN remaining.
Preprocessing — X-Ray	✓ DONE	Feb 2026	224×224, [0,1], augmentation configured, 70/15/15 split.
Data loaders validated	✓ DONE	Feb 2026	TF ImageDataGenerator <0.5s/batch confirmed.
Branch A training — 7 models	✓ DONE	Feb 25, 2026	Best: Gradient Boost AUC=0.9960. 13 charts. Models saved.
Branch B training — 8 CNNs	⌚ NOT STARTED	March 2026	8 architectures with 2-phase transfer learning. Next step.
Late fusion implementation	⌚ DESIGNED — code ready	After Branch B	Code written. Awaits CNN AUC weights for weighted average.
Full multimodal evaluation	⌚ PLANNED	March 2026	ROC, SHAP, Grad-CAM after both branches complete.

SECTION 7 — Self-Assessment & Grading Rubric

Criterion	Max Points	Self-Assessment	Instructor Score	Comments
1. Dataset Understanding & EDA	20	19/20 — Both datasets fully described. Age mismatch identified. Feature inventory with actual training importances confirmed by results.		
2. Issue Identification	15	14/15 — 10 issues found, rated, and fixed. Two critical issues resolved.		
3. Preprocessing Strategy	20	19/20 — Complete pipeline with full code for both branches. sklearn Pipeline confirms zero leakage. CSV pipeline executed. X-Ray pipeline validated.		

Criterion	Max Points	Self-Assessment	Instructor Score	Comments
4. Class Imbalance Handling	15	14/15 — Class weights computed with formula. CSV: class_weight='balanced' in 7 models. X-Ray: 1.850/0.685 ready. Augmentation configured. SMOTE rejection clinically justified.		
5. Augmentation (if applicable)	10	9/10 — 6 transforms with medical justification. V-Flip correctly excluded. Train-only application confirmed.		
6. Pipeline Implementation	10	9/10 — CSV pipeline fully executed: 0 NaN, 7 models, 13 charts, models saved to GitHub. X-Ray pipeline validated but CNN training not yet started.		
7. Documentation Quality	10	9/10 — All sections complete, actual results included, full code provided, decisions justified clinically.		
TOTAL	100	93 / 100 (self-estimated)		

In addition to the preprocessing code shown above, the X-Ray branch includes comprehensive technical documentation and deliverables that form the foundation for model training:

Detailed Technical Reports (GitHub):

1. Data Exploration Report (DATA_EXPLORATION_REPORT.md)

- Complete analysis of 5,856 pediatric chest X-rays
- Class imbalance identification (2.7:1 ratio PNEUMONIA:NORMAL)
- Image quality analysis with Laplacian variance thresholds (NORMAL <87.7, PNEUMONIA <62.3)
- Variable dimension analysis (384×127 to 2,916×2,713 pixels)
- Figures 01-04: Distribution charts, sample images, dimension histograms, pixel analysis

2. Data Preprocessing Report (DATA_PREPROCESSING_REPORT.md)

- Dataset reorganization with stratified 70/15/15 split
- **CRITICAL FIX:** Validation set increased from 16 → 878 images (54× larger)
- Class weights calculation via sklearn: NORMAL (1.850) / PNEUMONIA (0.685)
- Intelligent augmentation pipeline design (rotation ±15°, no vertical flip for anatomical correctness)
- Preprocessing pipeline: resize 224×224, normalize [0-1], grayscale conversion
- Figures 05-07: Quality analysis, augmentation examples, pipeline visualization

3. Data Loaders Report (DATA_LOADERS_REPORT.md)

- TensorFlow ImageDataGenerator configuration with optimal parameters
- Performance optimization: <0.5 seconds per batch on T4 GPU
- Batch validation: correct shape (32, 224, 224, 1), pixel range [0.0, 1.0], float32 type
- Reusable Python package: data_loader.py for seamless ML Engineer integration
- Figures 08-09: Batch examples with augmentation, distribution validation across splits

Critical Achievements:

✅ **Validation Set Crisis Resolved:** Original Kaggle validation set contained only 16 images (8 NORMAL + 8 PNEUMONIA). With 8 CNN architectures to compare, one misclassified image = 6.25% accuracy swing, making early stopping and model selection completely unreliable. Solution: Rebuilt entire split using stratified sampling → validation set grew to 878 images (54× increase), enabling robust model selection.

✓ **Class Imbalance Strategy:** Calculated optimal class weights using sklearn `compute_class_weight('balanced')` → {0: 1.850, 1: 0.685}. This forces the model to give 2.7× more attention to the minority NORMAL class during training, compensating for the severe 2.7:1 imbalance without data duplication or synthetic image generation (SMOTE rejected as clinically inappropriate for medical images).

✓ **Anatomically-Correct Augmentation:** Designed augmentation pipeline with medical domain constraints: rotation ±15° (realistic patient positioning), horizontal flip enabled (lungs are symmetric), but vertical flip DISABLED (heart position matters, anatomically incorrect if flipped). Applied ONLY to training set — validation and test sets receive normalization only.

✓ **Performance Optimization:** Achieved <0.5 seconds per batch loading time on Google Colab T4 GPU through optimal ImageDataGenerator configuration, enabling efficient training across 8 CNN architectures without I/O bottlenecks.

Complete Deliverables Package:

- Preprocessed dataset: 4,099 training / 878 validation / 879 test images
- 3 comprehensive technical reports with complete methodology documentation
- 9 professional visualization figures (01_class_distribution.png through 09_loader_distribution.png)
- Configuration files: preprocessing_config.json, data_loaders_summary.json
- Reusable Python package: data_loader.py with PneumoniaDataLoader class
- Performance validated: batch shape (32,224,224,1) ✓, pixel range [0,1] ✓, float32 ✓, loading time <0.5s ✓

Student comments / questions for the instructor:

Q1 — CSV training COMPLETE: Branch A is fully trained. Best model: Gradient Boost — AUC=0.9960, Accuracy=93.8%, Precision=1.000 (zero false positives), Recall=0.857, F1=0.9231, CV AUC=0.9939±0.009. All 7 models evaluated with 5-fold CV. 13 charts, all models saved to GitHub. Code executes smoothly end-to-end on Google Colab.

Q2 — Code execution: pediatric_pneumonia_analysis_COLAB.ipynb runs smoothly cells 0→48 without errors. Steps: Drive mount → GitHub clone → libraries → load CSV → preprocessing → EDA (8 charts) → 7 model training → evaluation (5 charts) → save models → push to GitHub. Zero errors and the same for other code.

Q3 — Age filter: We used ages 1–16 (127 patients) rather than strict 1–5 to preserve training samples. Please advise if strict 1–5 is required despite smaller sample size.

Q4 — CSV validation: We use 5-fold CV (no fixed val set) for CSV branch. Validation exists dynamically inside each CV fold (~19 patients per fold). Please confirm this is acceptable for n=127.

Students' Signatures

1. Kassouar Fatima (Project Manager · CSV · ML · Report)

Signature: _____

2. Bouhmidi Amina Maroua (Data Engineer · X-Ray)

Signature: _____

3. Labani Nabila Nour Elhouda (DL Engineer · X-Ray)

Signature: _____

4. Miloudi Meroua Amira (Developer · Business)

Signature: _____

Instructor Signature

Dr. Abderrahmane Khiat

Signature: _____

Date: _____

Grade: _____ / 100

